PKM prime fall degree, stage C and gate G4 (job pfd-c-20261108)

Every number is [pcc:unverified]. Source: research/pkm_falldegree_unblock_20261107/results/C9/3eeef6a6-9565-415c-918c-b3f92801aafc/ (job pfd-c-20261108). Scored by the scratchpad scorer pkm_c_score.py (archived, scripts.tar.gz); cross-checked against the committed prime_fall_degree_tools score (dev build of unchanged source): 44 cells, 0 median mismatches, 0 H-m mismatches.

Verdicts (preregistration §5 C, §6, Amendments 1, 3, 4, 7 C, 10, 11 as written)

H-m (fixed t, ladder C): GROWS IN m on all six ladders (P1 Kummer t=2,3; P0 Kummer t=2,3; P0 2-isogeny t=2,3). The reading uses completed (exact) medians only and does not depend on any m >= 7 row (closure values; stage C G3 passed on m <= 6). Without the small-base cells, P0 isogeny t=2 would read 'mixed' (5,6,6); §6 says small-base cells enter the rule, so the registered reading is 'grows'.

kummer P1 t=2: m=2..10 d_F medians 4 5 6 6 >=7 >=8 >=8 >=8 >=7 -> H-m: GROWS (m<=6 only: grows; without small base: grows; ffd medians 4 4 4 4 4 4 4 4 4)

kummer P1 t=3: m=2..10 d_F medians 4 6 7 >=8 >=8 >=8 >=7 >=6 >=6 -> H-m: GROWS (m<=6 only: grows; without small base: grows; ffd medians 4 4 4 4 4 4 4 4 4)

isogeny P0 t=2: m=2..10 d_F medians 5 6 6 7 >=7 >=8 >=8 >=8 >=7 -> H-m: GROWS (m<=6 only: grows; without small base: mixed; ffd medians 5 5 5 5 5 5 5 5 5)

isogeny P0 t=3: m=2..5 d_F medians 5 6 7 >=8 -> H-m: GROWS (m<=6 only: grows; without small base: grows; ffd medians 5 5 5 5)

kummer P0 t=2: m=2..10 d_F medians 4 5 6 6 >=7 >=8 >=8 >=8 >=7 -> H-m: GROWS (m<=6 only: grows; without small base: grows; ffd medians 4 4 4 4 4 4 4 4 4)

kummer P0 t=3: m=2..5 d_F medians 4 6 7 >=8 -> H-m: GROWS (m<=6 only: grows; without small base: grows; ffd medians 4 4 4 4)

H-LFD: m=2 stage C systems repeat stage A rows exactly (12 of 12 draws agree on x_r, curve, d_F, ffd, falls, muladds): no change. m=3..6: pending stage B (pfd-b), which owns those ladders. m>=7: no completed N, no reading.

H-FFD, H-gap, G2: not applicable. No D row exists for any stage C cell (run.sh has no d-C stage; Amendment 3's DC-* rows were never run).

H-null: no null arm in stage C. Amendment 6/7/8 readings: no stage C cell enters them.

Amendment 10 (high lambda >= 0.1, m >= 3): no stage C cell (max lambda 5.6e-3, P0 t=3 m=5). Amendment 7 C 'possibly decomposable' (lambda >= 0.01): none. Kummer m=2 draws: grid_zeros 0 (stage A certificate, same systems).

Gates

G1: PASS, 6 hand cases ('G1: 6 cases pass').

G4 (planted): PASS. 32 planted rows over 16 cells (Kummer P1 m=2 t=1..6, m=3 t=2,3; isogeny P0 same), 2 draws each; closure to cmax 5: 1 never enters (0 rows with 1), planted point solves every generator on all rows (planted_ok false/null: 0), 0 censored. Stage A's U4 condition is met.

G3: not run (tool defect) in pfd-c (Amendment 11: gate3 launched capped F4 at m >= 7, pilot panicked at f4_fp_tower.rs:736, exit 101, empty g3.jsonl). Stage C G3: PASS on the m <= 6 cells in job pfd-g3c-20261109: 'G3: 98 capped runs, 0 disagreements', exit 0; re-derived from g3.jsonl: 98 runs, agree 98, timeouts 0, 56 systems in 28 cells, equal to the Amendment 11 within-capacity set (0 missing, 0 extra); inputs 19/19 hashes match pfd-c. m >= 7: G3 not applicable. Amendment 11 stop rule: the study continues.

Instrument

31/31 registered run files, 44/44 cells, 88 draws, two per cell (k=0,1); argv in progress.txt equals run.sh hky-C/g4 for every stage; seed 1347112276 on every summary line; no duplicate, missing or extra draw; x_r distinct.

Censoring (§2), draws: exact 42; cmax 7 reached without 1 (d_F > 7) 28; limit-stopped (d_F >= c): budget 4, max_entries 2, max_cols 12.

m >= 7 block (Amendment 11)

Labelled 'G3 not applicable (F4 engine capacity)'. t=2 (P1 Kummer, P0 Kummer, P0 isogeny): m=7,8,9 d_F > 7; m=10 d_F >= 7 (max_cols). t=3 (P1 Kummer only): m=7 d_F > 7; m=8 >= 7; m=9,10 >= 6 (max_cols). No gate checks the closure at 5-8 free variables (G1 cases use 0, 1, 3; G3 reaches 4; G4 runs at m = 2, 3). These rows enter H-m but the reading does not depend on them.

Execution

Job pfd-c-20261108 (Amendment 9: 12 vCPU, RAYON 12, Xeon 8488C, 64 GB), pod ran 22:50:08Z-02:38:38Z; g1, the four g4 runs and all 31 hky-C closures exit 0; g3 exit 101; pod exit 1 (EKS audit log). Salvaged by card ...-pfd-c-C9-F01: 76 files, local sha256 list verified (76 OK).

No outcome is an S, a price or a leaderboard row (§8). Toy primes, t = 2, 3, m <= 10, two draws per cell (equal in every cell).

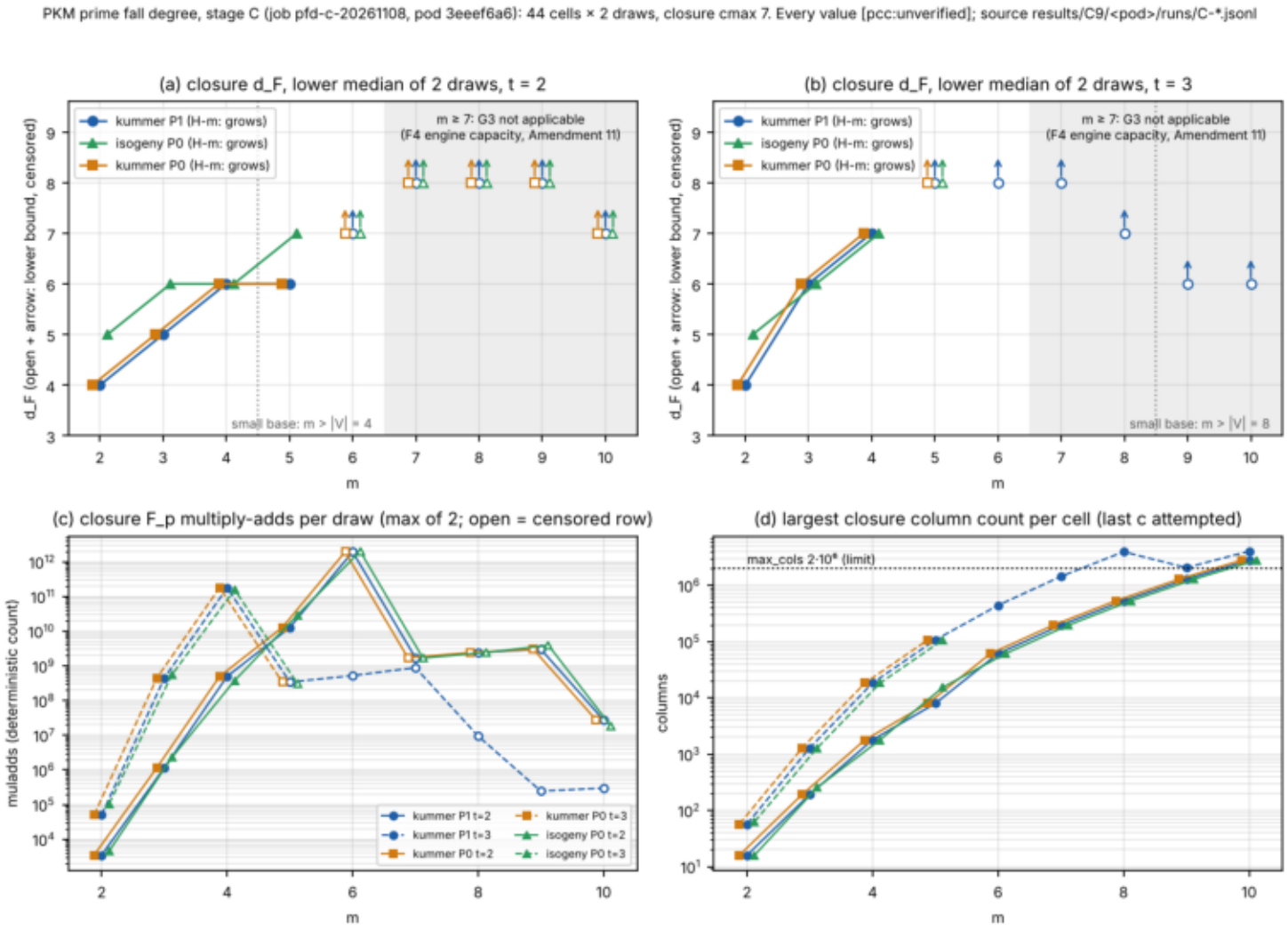


Figure 1 (results/C9/analysis/pfdC_fig1.svg). (a, b) lower-median closure d_F against m at t = 2, 3; open markers with arrows are censored lower bounds; grey: m >= 7 (G3 not applicable); dotted: small base m > |V| = 2^t. (c) closure F_p multiply-adds per draw. (d) largest closure column count. Recorded values only.